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A Literature Survey on Embryo Classification in Assisted Reproductive Technology (ART)

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Abstract

Embryo classification plays a crucial role in improving the success rates of in vitro fertilization (IVF). Automated assessment techniques leveraging computer vision and machine learning have emerged to support embryologists in selecting viable embryos. This literature survey reviews traditional morphological grading systems, machine learning classifiers, and deep learning-based embryo evaluation methods. The paper also outlines key challenges and future prospects in automating embryo quality assessment.

Keywords: Embryo classification · IVF · Deep learning · Morphology · Time-lapse imaging · Artificial intelligence

1 Introduction

Infertility affects millions globally, and IVF is one of the most effective treatments. A critical step in IVF is selecting the embryo **most likely to result in a successful pregnancy**. Traditionally, **this** decision is made based on visual assessment of embryo morphology and development patterns. However, this process is subjective and highly dependent on the embryologist's expertise.

Automated embryo classification systems aim to provide objective, reproducible, and scalable methods for embryo selection. This survey reviews the evolution of such systems, highlighting key advances in image analysis and AI-driven classification.

2 Related Work

Traditional Grading Techniques

Embryologists use criteria such as blastomere number, fragmentation percentage, and cell symmetry to score embryos manually. Despite being widely practiced, these methods lack reproducibility.

Machine Learning Approaches

Handcrafted Feature-Based Classification: Early computational approaches extracted features like texture, cell boundary shape, and developmental time points. Support vector machines (SVM), decision trees, and logistic regression were commonly used for classification [1].

Deep Learning Techniques

CNNs for Static Image Classification: Convolutional Neural Networks (CNNs) have been used to classify embryo images at various development stages, achieving superior accuracy over traditional methods [2].

Time-Lapse and Video-Based Models: Time-lapse incubators enable dynamic embryo monitoring. Deep learning models, particularly LSTM and 3D CNNs, have been applied to analyze spatiotemporal patterns for predicting implantation potential [3].

Multimodal and End-to-End Models: Recent approaches combine image features with patient metadata (e.g., age, hormone levels) in multimodal networks, improving prediction performance [4].

3 Challenges and Future Work

Automating embryo classification remains challenging due to:

- Limited and non-standardized datasets
- Ethical and regulatory concerns about AI in reproductive medicine
- Label noise and subjectivity in ground-truth annotations
- Generalization across clinics and imaging devices

Future work should focus on explainable AI, federated learning, and clinical validation of automated systems to support their adoption in real-world IVF labs.

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